

Medication Therapy Management and Complex Patients With Disability: A Randomized Controlled Trial

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Elizabeth A. Chrischilles, PhD¹, William Doucette, PhD¹, Karen Farris, PhD², Scott Lindgren, PhD¹, Brian Gryzlak, MSW¹, Linda Rubenstein, PhD¹, Kelly Youland, PharmD³, and Robert B. Wallace, MD, MS¹

Abstract

Background: Drug therapy problems, adverse drug events (ADEs), and symptom burden are high among adults with disabilities. **Objective:** To compare the effects of a modified medication therapy management (MTM) program within a self-efficacy workshop versus the workshop alone or usual care on symptom burden among adults with activity limitations. **Methods:** Three-group randomized controlled trial among adults (age 40 and older) with self-reported activity limitations in community practice. Interventions: 8 weekly Living Well With a Disability (LWD) 2-hour workshop sessions with and without a collaborative medication management (CMM) module. Primary outcome: mean number of moderate to very severe symptoms from a list of 11 physical and mental symptoms. Process measures: changes in medication regimens and self-reported ADEs. Analysis: general linear mixed models (continuous outcomes) and generalized estimating equations (categorical outcomes). **Results:** Participants had high symptom burden, low physical health, and took many medications. There was a significant increase in ADE reporting in the LWD + CMM group relative to the other 2 groups (Study group $\times$ Time $P = .014$), and there were significantly more changes in medication regimens in the LWD + CMM group ($P = .013$ LWD only vs LWD + CMM). The oldest third of participants had significantly fewer mean symptoms but received more intense CMM. There was no difference between the LWD-only, LWD + CMM, and usual care groups in symptom burden over time. **Conclusion:** Pharmacist MTM practices and MTM guidelines may need to be modified to affect symptom burden in a population with physical activity limitations.

Keywords

pharmacists, persons with disabilities, symptom evaluation

Introduction

One in 5 Americans has a disability (Centers for Disease Control and Prevention, 2009), and people with disability face a 3.5 times higher health care cost than those without a disability.^{1,2} People with disabilities experience a high burden of symptomatic clinical problems related to their underlying disability, and these problems are often described as “secondary conditions.”³ In a recent study, the cumulative symptom frequency explained half of the total effect of disability on general health status and about one-third of the total effect of disability on physical functioning.³ Improvement in pharmacotherapy may be a way to reduce some of this symptom burden. By causing side effects, medication use may aggravate common symptoms such as fatigue, dizziness, dry mouth, or stomach problems. Suboptimal medication use may contribute to poor

management of pain, which is the most prevalent of all secondary conditions.⁴

Shortly before the Medicare Part D medication therapy management (MTM) program went into effect, we evaluated a model of MTM for complex Medicaid patients (taking 4 or more medications for chronic conditions). Using what was called pharmaceutical case management (PCM) at the time, we found that community pharmacists improved

¹University of Iowa, Iowa City, IA, USA

²University of Michigan, Ann Arbor, MI, USA

³Gundersen Lutheran Medical Center, LaCrosse, WI, USA

Corresponding Author:

Elizabeth A. Chrischilles, PhD, Department of Epidemiology, College of Public Health, University of Iowa—Iowa City, 5424 CPHB, 105 River Street, IA 52242, USA.

Email: e-chrischilles@uiowa.edu

Table 1. List of Study Activities by Study Group Assignment.

Activity	Study Group		
	LWD Only	LWD + CMM	Comparison Group
Attend LWD group sessions (2 hours, once a week, for 8 weeks)	√	√	
Attend a group medication management session (one 2-hour session, cofacilitated by study pharmacist)		√	
Meet individually with a study pharmacist (typically 4 times)		√	
Complete 3 telephone interviews (after enrollment, then 6 and 12 months later)	√	√	√
Consent to medical and pharmacy records release	√	√	√
Receive \$100 reimbursement	√	√	√

Abbreviations: LWD, living well with a disability; CMM, collaborative medication management.

medication appropriateness and successfully reduced the use of potentially inappropriate medications among elderly people.⁵ In a separate study, we learned that adults with physical disability (mobility limitations) had a high prevalence of adverse drug events (ADEs)⁶ and inappropriate medication use.⁷ We subsequently conducted the present study to evaluate whether a modified version of the PCM intervention (collaborative medication management, CMM), when combined with a self-management program designed to reduce secondary conditions for adults with disabilities (Living Well With a Disability, LWD), could reduce symptom burden for patients with self-reported activity limitations. The LWD program has been shown to improve limitation from secondary conditions in a randomized wait-list control trial conducted in Centers for Independent Living (CIL) in 8 states.⁸

We theorized that overall symptom burden, whether attributable to ADEs, disability, or secondary conditions, should be amenable to reduction through modification of pharmacotherapy—either through remediation of ADEs or symptom undertreatment (eg, pain management). We hypothesized that the complementary mechanisms of the 2 interventions would result in greater reduction in symptom burden than the Living Well program alone. The combination of CMM with the LWD program constitutes a new, integrated intervention that has not previously been tested. The aim of this study was to compare the 6- and 12-month mean symptom prevalence among adults with self-reported activity limitations randomized to 1 of 3 groups: (1) LWD program augmented with CMM, (2) traditional LWD program, or (3) no intervention (usual care).

Methods

Study Population and Study Design

A screening questionnaire was mailed to an age-stratified random sample of 34 804 community-dwelling adults drawn from the voter registration lists for the 3 Iowa

metropolitan areas (Cedar Rapids, Des Moines, and Dubuque), and 12 330 (35.4%) responded. Eligible individuals were those who answered “limited a lot” to at least 1 of 9 questions about activity limitations in the questionnaire, reported taking ≥ 3 medications, responded “fairly often” or “very often” to any of a list of common physical and mental symptoms, and indicated that they did not have plans to move out of the area in the next year. The activity limitation items were 9 of the 10 physical function items (minus 1 item about limitation in vigorous activity) from the Medical Outcomes Study 36-item Short-Form (SF-36).⁹ All eligible individuals ($n = 2071$) were mailed an invitation to participate in the intervention trial, and those who agreed to participate completed a telephone interview (baseline survey) and were informed of their study group. The target enrollment was achieved, and a total of 305 people were randomized to 3 groups. Randomization occurred in 3 waves over 3 years to accommodate the desirable class sizes, which placed an upper limit on the number of participants per year for the intervention group programs. Within each site, randomization was conducted using block randomization with a block size of 3 to ensure that equal numbers were assigned to each of the 3 study groups. We randomized 3 people at a time using sealed envelopes that contained the assignment order that had been randomly pre-assigned by computer. The envelopes were prepared by an individual not involved in the interventions or data collection. In addition to the baseline survey, 2 annual follow-up telephone surveys were conducted to collect all study measures.

Study Interventions

In Table 1, we display the activities involved in each of the 3 study groups. The LWD program^{8,10-13} consists of 8 weekly 2-hour in-person workshops focused on self-management and health promotion for people with disabilities. Transportation to the sessions was provided when necessary. The sessions took place at a CIL, a community health

facility (YMCA), a community hospital, and a local Arc chapter (Arc is a national community-based organization advocating for and serving people with intellectual and developmental disabilities) in the study cities. A certified LWD facilitator (<http://www.livingandworkingwell.org>) led the discussion-based workshop sessions. Participants received a copy of the LWD handouts that include assignments for each topic. Topics include goal setting, problem solving, attribution training, managing depression, communication, information seeking, nutrition, physical activity, and advocacy. The first 4 topics establish goal pursuit, and the final 5 topics encourage health behavior change as objectives to meeting quality-of-life goals.

The LWD + CMM intervention consisted of a workshop session on medication management added to the LWD program plus a goal of 4 one-on-one in-person or telephone visits with a study CMM pharmacist over 9 months. Session participants received the LWD handouts and a separate CMM workbook developed for the workshop. The new workshop session emphasized important medication management strategies (keeping a medication list, knowing what questions to ask, etc) and focused on patient role and expectations. The session was developed and pretested with input from an external advisory committee of people with physical disabilities and is similar to a booklet developed for older adults.¹⁴ For the one-on-one sessions, we identified 1 community pharmacist, already trained in MTM methods as part of their doctor of pharmacy training, from each of the 3 study sites who provided CMM services to study participants. Two of the community pharmacists were practicing in retail settings and one in a clinic setting. Pharmacists were also trained in a 2-hour session and periodic booster sessions to realize the symptom burden experienced by those with disabilities and to understand that the purpose of the study was to reduce symptoms from either ADEs or untreated conditions. Study pharmacists also attended the CMM workshop session to meet study participants and schedule the first one-on-one visit.

The 3 pharmacists were instructed to provide MTM consistent with the pharmacy profession's consensus model for MTM and that used under PCM: a comprehensive medication therapy review, a personal medication record, a patient medication-related action plan, an intervention to implement the plan (such as patient education, contact with a prescriber, and/or a referral), documentation, and follow-up. Pharmacists documented these visits in patient charts kept in the pharmacy, and physicians were contacted for follow-up or to recommend a change. Documentation included a description of the problems identified, the recommended actions, and whether the physician authorized the pharmacist to implement the changes. Pharmacists communicated with the physician via fax to make recommendations for changes in

medication therapy. Copies of these faxes were kept in the patient charts. Physicians could respond to faxed recommendations by signing the space on the fax documenting their approval of the recommendation, or they could write modifications to the pharmacist's recommendation. These signed faxes are legal prescriptions in the state of Iowa. Because many participants were not usual patients of the study pharmacists, the study pharmacists also communicated with the patient's dispensing pharmacy regarding any changes in medication therapy.

The comparison group received neither LWD nor CMM, but participants in this group completed all 3 study surveys. The LWD intervention had only been evaluated in 1 prior randomized wait-list control study,⁸ and it was important to confirm the positive effects in our study population by including a no-intervention comparison group.

Measures

The primary outcome variable was the mean number of 11 self-reported symptoms rated as moderate, severe, or very severe that had occurred on more than 7 days in the past month. The symptom checklist was derived from symptoms included in a published measure of ADEs¹⁵ and in a measure of secondary conditions that affect people with disabilities.⁴ Secondary study outcomes included prevalence of each moderate or severe symptom; response (yes/no) to the question, "In the past 6 months have you had any side effects, unwanted reactions, or other health problems from medications you were taking?"; mean number of medications; and the calculated medication activity (sum of medications started and medications stopped over the year). A complete inventory of prescription medications was collected at each survey. Participants were asked to collect their medication bottles and read from the labels whenever possible.

Other measures from the baseline survey were used to describe the study sample and assess balance achieved through randomization (Table 2). These included age, race (white and other), mean number of 10 comorbidities, indicators for each comorbidity, 7 of the 8 health-related quality-of-life scales from the RAND 36-item SF-36,⁹ and 1 item from the 5-item SF-36 general health scale ("Compared to others your age would you say your health is excellent, very good, good, fair, or poor").

Statistical Methods

Primary analyses involved intention-to-treat analyses of change over time (baseline, BL; 6-month follow-up, F6; and 12-month follow-up, F12) for the 3 study groups (LWD, CMM + LWD, and no intervention). A generalized linear mixed model that allowed for nonconstant variance, correlation between observations on the same subject, and

Table 2. Baseline Characteristics of Study Participants, Overall and by Study Group (n = 294).

Characteristics at Baseline	Study Group				Group Difference P Value ^{a,b}
	All (n = 294)	LWD Only (n = 97)	LWD + CMM (n = 100)	No Intervention (n = 97)	
Mean (SE) age	65.1 (0.7)	64.4 (0.5)	64.5 (0.5)	64.4 (0.5)	.994
Age group (years), n (%)					.625
42-59	106 (36.0)	40 (41.2)	33 (33.0)	33 (34.0)	
60-69	72 (24.5)	24 (24.7)	25 (25.0)	23 (23.7)	
≥70	116 (39.5)	33 (34.0)	42 (42.0)	41 (42.3)	
Gender n (%)					.953
Female	195 (66.3)	66 (68.0)	66 (66.0)	63 (65.0)	
Male	99 (33.7)	31 (32.0)	34 (34.0)	34 (35.0)	
Race, n (%)					.128
White	267 (91.1)	89 (92.7)	86 (86.0)	92 (94.8)	
Other	26 (8.9)	7 (7.3)	14 (14.0)	5 (5.2)	
Enrollment site, n (%)					.873
Cedar Rapids	96 (32.7)	35 (36.1)	33 (33.0)	28 (28.9)	
Des Moines	97 (33.0)	30 (30.9)	34 (34.0)	33 (34.0)	
Dubuque	101 (34.3)	32 (33.0)	33 (33.0)	36 (37.1)	
Enrollment wave, n (%)					.816
One	103 (35.0)	32 (33.0)	36 (36.0)	35 (36.1)	
Two	80 (27.2)	29 (29.9)	28 (28.0)	23 (23.7)	
Three	111 (37.8)	36 (37.1)	36 (36.0)	39 (40.2)	
Single-item health status, n (%)					.211
Excellent/Very good	42 (14.3)	16 (16.5)	11 (11.1)	15 (15.5)	
Good	98 (33.5)	37 (38.1)	33 (32.3)	29 (29.9)	
Fair/Poor	153 (52.2)	44 (45.4)	56 (56.6)	55 (54.5)	
Quality of life means (SE)					
Physical function	37.6 (1.7)	37.5 (2.0)	38.3 (2.0)	38.2 (2.0)	.862
Physical role	38.3 (1.5)	37.0 (2.6)	35.7 (2.6)	40.1 (2.6)	.450
Vitality	43.5 (1.3)	44.0 (2.2)	41.5 (2.1)	46.0 (2.2)	.322
Bodily pain	46.7 (1.2)	48.5 (2.1)	45.8 (2.1)	47.7 (2.1)	.631
Social role	63.0 (1.8)	60.2 (3.1)	62.3 (3.0)	64.4 (3.1)	.616
Emotional role	80.1 (1.4)	78.5 (2.4)	81.1 (2.4)	80.6 (2.4)	.711
Mental	64.0 (1.4)	65.3 (2.3)	62.2 (2.4)	65.3 (2.4)	.537
Mean (SE) comorbid conditions (SE)	3.4 (0.1)	3.5 (0.2)	3.4 (0.2)	3.4 (0.2)	.943
Comorbid conditions, n (%)					
Anemia	59 (20.4)	23 (24.2)	20 (20.4)	16 (16.7)	.389
Arthritis	224 (76.5)	80 (82.5)	73 (73.7)	71 (73.2)	.214
Cancer	68 (23.5)	24 (25.3)	26 (26.3)	18 (18.7)	.330
Diabetes	101 (34.4)	30 (30.9)	40 (40.0)	31 (32.0)	.309
Emphysema	116 (39.6)	43 (44.3)	36 (36.0)	37 (38.5)	.818
Heart attack	48 (16.4)	16 (16.7)	15 (15.2)	17 (17.5)	.760
Hypertension	116 (72.1)	62 (65.3)	73 (73.7)	75 (77.3)	.158
Liver disease	13 (4.5)	4 (4.1)	6 (6.1)	3 (3.1)	.666
Stroke	65 (22.3)	22 (23.2)	19 (19.0)	24 (25.0)	.552
Ulcers	104 (35.9)	32 (34.0)	35 (35.0)	37 (38.5)	.848
Mean (SE) symptoms ^c	2.9 (0.1)	3.1 (0.2)	3.0 (0.2)	2.5 (0.2)	.077
Symptom, n (%)					
Breathing difficulties	58 (19.7)	20 (20.6)	15 (15.0)	23 (23.7)	.100
Feeling confused/trouble concentrating	42 (14.3)	20 (20.6)	14 (14.0)	8 (8.2)	.062
Depressed/low mood	61 (20.7)	19 (19.6)	23 (23.0)	19 (19.6)	.628
Dizziness, lightheadedness, or problems with balance	37 (12.6)	15 (15.5)	13 (13.0)	9 (9.3)	.742
Dry mouth/blurry vision	101 (34.4)	38 (39.2)	36 (36.0)	27 (27.8)	.371
Extreme fatigue/persistent tiredness	118 (40.1)	42 (43.3)	42 (42.0)	34 (35.1)	.438
Leaking of urine/problems urinating	55 (18.7)	18 (18.6)	22 (22.0)	15 (15.5)	.351
Pain	179 (60.9)	62 (63.9)	64 (64.0)	53 (54.6)	.210
Skin problems (rashes, itching, burning, unusual bruising, pressure sores, skin ulcers)	46 (15.6)	16 (16.5)	16 (16.0)	14 (14.4)	.643
Problems with sleep	100 (34.0)	41 (42.3)	34 (34.0)	25 (25.8)	.170
Stomach/bowel difficulties (eg, nausea, vomiting, abdominal pain, constipation, diarrhea)	62 (21.1)	24 (24.7)	24 (24.0)	14 (14.4)	.037
Mean (SE) number of drugs	8.5 (0.2)	8.0 (0.4)	9.1 (0.4)	8.3 (0.4)	.165
Range	2-28	3-19	2-27	3-28	

Abbreviations: LWD, living well with a disability; CMM, collaborative medication management; SE, standard error; NA, not applicable.

^aAdjusted for age, gender, site, and wave.^bCochran-Mantel-Haenszel general association P value for categorical variables and adjusted least-squares means P values for mean differences.^cModerate or severe symptoms lasting >7 days in the past month.

missing data points was used to assess key continuous outcome measures across time points (surveys). Categorical data models utilized generalized estimating equations (GEE) methodology. GEE models were used with a logit link to examine change over time. These GEE methods also allowed correlation between observations on the same subject and missing data points. All models adjusted for age, gender, study site, and study enrollment wave. All models included the multiplicative interaction between survey and group of interest (eg, age group and treatment group) to assess differences in change over time. All statistical analyses were performed using SAS version 9.2 (SAS Institute, Carey, NC). Our study sample size was planned to detect a clinically meaningful difference between groups in mean symptoms of 2.0 with at least 80% power.

Exploratory analyses were also conducted of 4 post hoc “as-treated” groups (no treatment, LWD only, CMM only, and LWD and CMM). The post hoc treatment groups were defined based on attendance by participants at their assigned study group sessions: participants in the intervention groups who did not actually participate in an intervention were assigned to the post hoc “no treatment” group along with participants randomized to this group. Those who participated in at least 4 of the 8 sessions were considered to have received LWD. Participants who had at least 1 visit with a CMM pharmacist were considered to have received CMM. Exploratory subgroup analyses over time were also performed by gender, age groups, race, and health status groups.

The protocol was approved by the University of Iowa Institutional Review Board for the Protection of Human Subjects, and signed informed consent was obtained from all study participants.

Results

Approximately three-fourths of the participants completed the full intervention to which they were assigned: 69 (71%) of the LWD group and 76 (76%) of the LWD + CMM group. An additional 17 (17%) from the LWD + CMM group participated in only the CMM component. Among the LWD group participants, the mean number of LWD sessions was 5.1 (standard error [SE] = 0.3, range = 0-8). The LWD + CMM group participants participated in a similar number of LWD sessions (mean = 5.4; SE = 0.3; range = 0-8). The CMM workshop was attended by 68% of the LWD + CMM group participants. More than 92% (n = 61) of those attending the workshop had at least 1 additional contact with a pharmacist, with a mean of 2.5 pharmacist visits (SE = 0.2; range = 0-5), whereas 78% (n = 25) of those not attending the workshop had at least 1 additional contact and 2.4 (SE = 0.1) mean visits (range = 0-5). Counting the workshop, the LWD + CMM participants had 3.1 CMM contacts (SE = 0.2; range = 0-6), with 84% having 2 or more contacts.

Table 3. Drug Therapy Problems Identified During CMM.

Drug Therapy Problem	Frequency (%)
Needs additional therapy	59 (31.1)
Adverse drug reaction	24 (12.6)
Inappropriate adherence	20 (10.5)
Dosage too low	20 (10.5)
Unnecessary drug therapy	19 (10.0)
Cost	18 (9.5)
No problem/issue	16 (8.4)
Wrong drug	7 (3.7)
Dosage too high	5 (2.6)
Drug interaction	2 (1.1)
Total	190 (100)

Abbreviation: CMM, collaborative medication management.

Completed 6- and 12-month follow-up interviews were conducted for 294 (96%) of the 305 who were randomized. The study groups were well balanced at baseline (Table 2). As expected, baseline physical health was below the population average. Mental health was above the population average.⁹ At baseline, participants took a mean of 8.5 (SE = 0.2; range = 2-28) medications and reported a 6-month ADE prevalence of 24.1%.

In Table 3, we display the types of drug therapy problems (DTPs) identified by the study pharmacists and in Table 4, the actions they took to resolve problems. The mean number of problems identified per patient was 1.9. The majority (60.5%) of pharmacist CMM actions involved communication with the physician regarding a change in medication regimen, and the most common drug class involved was cardiovascular (27.8%) followed by analgesic (23.3%). The most common change recommended was adding a drug. Of the 115 that communicated recommended changes to physicians, physicians agreed with recommended action on 49 (42.6%), disagreed 9 times (7.8%), and gave no response for 58 recommendations (50.4%).

There were no significant differences between the 3 study groups (ie, no significant Time × Study group interaction) in the primary study outcome (mean number of symptoms; Table 5). The mean number of symptoms declined slightly (0.2 symptoms for the LWD and no intervention group and 0.1 symptom for the LWD + CMM group) between the baseline and 12-month survey in all 3 groups (Table 5). There was a significant increase in ADE reporting in the LWD + CMM group relative to the other 2 groups (Study group × Time *P* value = .014), and there was significantly more medication activity in the LWD + CMM group (*P* = .013 LWD only vs LWD + CMM). When an as-treated analysis was conducted, results were unchanged (data not shown).

We additionally conducted analyses in subgroups we thought might have been especially likely to respond to

Table 4. Pharmacist Actions to Address Drug Therapy Problems (n = 190).

Pharmacist Action	n (%)
Recommend action to patient	40 (21.1)
Communicate with physician	115 (60.5)
Add drug	45 (23.7)
Switch drug	31 (16.3)
Increase dose	15 (7.9)
Discontinue drug	13 (6.8)
Decrease dose	9 (4.7)
Update status	2 (1.1)
Patient education	26 (13.7)
Patient monitoring	8 (4.2)
Referral	1 (0.5)

the interventions: those whose health status was less than very good and those in the 1 study site with the highest CMM activity. There were no significant differences between groups in the primary outcome in these subgroups. Though sample sizes were small and statistical significance not expected, the point estimates were not visually different.

We analyzed the relationship of CMM intensity to baseline level of symptoms (the primary study outcome) and age group. The proportion receiving high CMM intensity increased with age, but level of symptoms was inversely associated with age (Table 6).

Discussion

In this study among patients with physical activity limitations, patients at baseline had on average poor physical and self-rated health, were taking an average of 8.4 (SE = 0.3; range 2-28) medications, and reported a 6-month ADE prevalence of 24.1%. A novel model of pharmacist consultations combined with self-efficacy workshops resulted in increased awareness of ADEs and more changes in medications but no overall improvement in self-reported symptoms over the 12-month study. A study group that received the workshops but no pharmacist intervention also did not show an improvement in symptoms.

The CMM intervention increased ADE reporting, an important first step in order for corrective action to be taken. Messages in the workshop booklet and discussions reinforced awareness of ADEs, and the CMM group reported more ADEs. Although it is not possible to determine which component of the LWD + CMM intervention was responsible, these findings suggest that the booklet may be a useful component of MTM interventions. Designed in partnership with patients with physical disabilities, it may be of particular value for this vulnerable population.

Pharmacists recommended changes to medication regimens in the majority of cases, and more changes to medication regimens occurred in the group receiving pharmacist services. However, in spite of significantly increased patient awareness of ADEs and more medication changes indicating that pharmacists changed the process of care, there was no detectable effect of the intervention on symptom burden. There are several possible explanations for this unexpected finding. These include the patient population and their needs, validity of the primary outcome measure for this population, pharmacist clinical training, and whether organizational supports for care delivery were sufficient. Type 2 error is not a likely explanation because the amount of change over time was quantitatively as well as statistically similar (ie, decrease of 0.1 or 0.2 mean symptoms in all 3 study groups).

One possibility is that the study protocol may have required pharmacists to provide MTM to patients with lower need relative to previous studies. The number of DTPs (mean of 1.9) that pharmacists identified was toward the lower end of the range reported in prior studies, which have found 0.8 to 4.2 DTPs per patient annually.¹⁶⁻²¹ However, the study population was taking a large number of medications and had a high prevalence of ADEs relative to the general US population.^{22,23} In contrast to prior studies, pharmacists were required to meet with all study patients. In prior PCM studies, pharmacists had discretion in deciding to whom they would provide a service. It is possible that in those studies, pharmacists were able to identify a subset of patients with greater need for their services. Baseline data from one of those studies suggests that this might be true (in the Iowa Medicaid PCM program evaluation, there were a greater number of high-risk medications at baseline for eligible patients who received PCM vs eligible patients who did not).⁵ By requiring pharmacists to provide services to all study patients, some of whom may have had few or no DTPs, the impact of CMM could have been diluted. Finally, patients had to agree to be randomized; patients who choose to be involved in a study may be more involved with their own medical care and have fewer medication management problems as a result.

The primary end point was symptom burden. The interventions (either LWD alone or with CMM) may have affected domains of health not captured by the symptom checklist. There may have been an effect on symptoms, but the instruments may have been insensitive to those effects or the process changes may have truly had no effect on symptoms. The symptom checklist was derived from prior studies, but responsiveness of the instrument to change has not been established. Also, because the symptoms measured have many causes (some of which may not be amenable to medication management), the outcome

Table 5. Changes Over Time by Study Group.^a

Characteristic	LWD Only (n = 97)			LWD + CMM (n = 100)			No Intervention (n = 97)		
	Baseline	6 Months	12 Months	Baseline	6 Months	12 Months	Baseline	6 Months	12 Months
Primary outcome:									
Mean (SE) symptoms ^b	3.1 (0.2)	2.8 (0.2)	2.9 (0.2)	3.0 (0.2)	2.7 (0.2)	2.9 (0.2)	2.5 (0.2)	2.5 (0.2)	2.3 (0.2)
Secondary outcomes:									
Symptoms, n (%)									
Breathing difficulties	20 (20.6)	19 (19.6)	23 (23.7)	15 (15.0)	13 (13.0)	12 (12.0)	23 (23.7)	19 (19.6)	21 (21.6)
Feeling confused/ trouble concentrating	20 (20.6)	15 (15.5)	14 (14.4)	14 (14.0)	8 (8.0)	14 (14.0)	8 (8.2)	12 (12.4)	9 (9.3)
Depressed/low mood	19 (19.6)	23 (23.7)	18 (18.6)	23 (23.0)	18 (18.0)	19 (19.0)	19 (19.6)	15 (15.5)	15 (15.5)
Dizziness, lightheadedness, or problems with balance	15 (15.5)	18 (18.6)	13 (13.4)	13 (13.0)	13 (13.0)	15 (15.0)	9 (9.3)	8 (8.2)	9 (9.3)
Dry mouth/blurry vision	38 (39.2)	29 (29.9)	32 (33.0)	36 (36.0)	27 (27.0)	28 (28.0)	27 (27.8)	19 (19.6)	27 (27.8)
Extreme fatigue/ persistent tiredness	42 (43.3)	36 (37.1)	43 (44.3)	42 (42.0)	40 (40.0)	43 (43.0)	34 (35.1)	35 (36.1)	23 (23.7)
Leaking of urine/ problems urinating	18 (18.6)	19 (19.6)	21 (21.6)	22 (22.0)	10 (10.0)	16 (16.0)	15 (15.5)	15 (15.5)	12 (12.4)
Pain	62 (63.9)	57 (58.8)	65 (67.0)	64 (64.0)	67 (67.0)	67 (67.0)	53 (54.6)	55 (56.7)	55 (56.7)
Skin problems (rashes, itching, burning, unusual bruising, pressure sores, skin ulcers)	16 (16.5)	11 (11.3)	18 (18.6)	16 (16.0)	17 (17.0)	20 (20.0)	14 (14.4)	18 (18.6)	14 (14.4)
Problems with sleep	41 (42.3)	38 (39.2)	32 (33.0)	34 (34.0)	31 (31.0)	34 (34.0)	25 (25.8)	23 (23.7)	23 (23.7)
Stomach/bowel difficulties (eg, nausea, vomiting, abdominal pain, constipation, diarrhea)	24 (24.7)	23 (23.7)	21 (21.6)	24 (24.0)	24 (24.0)	24 (24.0)	14 (14.4)	19 (19.6)	15 (15.5)
Mean (SE) number of drugs	8.0 (0.4)	7.7 (0.4)	7.7 (0.4)	9.0 (0.4)	8.6 (0.4)	8.6 (0.4)	7.9 (0.4)	8.3 (0.4)	8.1 (0.4)
Mean (SE) medication changes ^c		6.9 (0.6)			8.5 (0.6)			7.9 (0.6)	
Adverse drug effects, n (%) ^d	27 (28.4)	17 (18.1)	10 (10.5)	20 (21.5)	29 (29.9)	27 (27.6)	21 (22.3)	19 (20.2)	10 (10.3)

Abbreviations: LWD, living well with a disability; CMM, collaborative medication management; SE, standard error.

^aAll analyses adjusted for survey, study group, sex, age group, site, and enrollment wave.

^bModerate or severe symptoms lasting >7 days in the past month.

^c $P = .0133$ LWD only vs LWD + CMM.

^dGroup $\times$ Time P value = .014.

Table 6. CMM Intensity and Mean Baseline Symptoms by Age, n = 294.

	Age Group (years)		
	<60 (n = 106)	60-69 (n = 72)	70+ (n = 116)
Number (%) with high CMM intensity ^a	10 (9.4)	13 (18.1)	25 (21.6)
Mean number of baseline symptoms ^b	3.9	3.0	1.8

Abbreviation: CMM, collaborative medication management.

^aHigh intensity was based on the following calculation: points were totaled for each patient problem and the types of actions the pharmacist took. The total was divided by the number of problems for that patient and added to a summary score based on attendance and pharmacist contact. The scores were then classified into none, low, moderate, and high CMM intensity based on percentiles; 50% were classified as receiving high CMM intensity.

^bBased on moderate or severe symptoms lasting 7 days or more.

measure may have been insensitive to pharmacist interventions. Proper conceptualization and measurement of the outcomes of MTM services has been previously noted as a knowledge gap in evaluating and improving MTM services.²⁴ A related possibility is that pharmacists may

not recognize or provide counseling for the symptomatic problems of people with physical disabilities. It was surprising to find that study pharmacists delivered the most intense CMM services to those with the fewest symptoms. Adding cardiovascular medications was the most common

recommendation, often a prevention-oriented action rather than a response to patient symptoms. Perhaps the CMM training could not overcome the mental schema developed through years of practice experience. Even with training to provide the CMM, pharmacists may have reverted to their regular PCM thinking, which likely focused on chronic medications, especially for cardiovascular conditions such as hypertension and hyperlipidemia. Thus, they may have provided PCM with a chronic condition focus rather than CMM with a symptom focus.

Study pharmacists also may have needed more work flow support. Perhaps they did not have the organizational supports and established work flow needed to deliver the care successfully. However, pharmacists were getting support from the research team through periodic telephone conferences to discuss barriers to implementing CMM; community pharmacists in previous studies did not have this support mechanism. Furthermore, one of the study pharmacists had an extremely well-developed work flow management system with demonstrated high follow-up rates. He shared his system, including communication methods, with the other study pharmacists, and regular conference calls attempted to keep motivation high. Results were not different for the pharmacist who developed the effective management plan compared with the other 2 pharmacists. It could also be that there were logistical barriers to novel aspects of the care model: the study did not address different communication needs and role expectations when introducing a study pharmacist into the existing patient-community pharmacist-physician triad. However, physicians authorized changes in therapy as a result of the CMM visits 42.6% of the time, which is close to results in other studies performed in the community setting (45%-50%).^{5,19,25}

A contributing factor to the null findings may also be that pharmacists need more time or information for new patients: they are used to working with patients with whom they have an existing relationship. In this study, the pharmacists were provided a medication list and symptom checklist that had been collected by the research team for each patient. They conducted a full medication history with the patient and provided patients with follow-up visits over a 9-month period. In prior PCM studies, community pharmacists delivered care to patients of their own pharmacy. Their existing relationship and long-term knowledge of patients' history of medication experiences may have improved their ability to identify DTPs.

Another factor influencing the group outcomes was the fact that the LWD intervention did not significantly reduce the frequency of moderate to severe symptoms. This is in contrast to a previous study of the LWD program conducted in CIL in 8 rural states.⁸ The outcome measure for that study used a similar list of symptoms, but respondents

rated the amount of limitation from each symptom on a 4-point scale (ranging from 0 = not a problem to 3 = limiting activity 11 or more hours per week). It is possible that LWD improves participants' ability to stay active without influencing frequency or severity of symptoms. In addition, the CIL population had a very different set of primary disabilities than our general population of people with activity limitations. Spinal cord injury (29.2%), arthritis (18.5%), cerebral palsy (16.2%), and multiple sclerosis (14.9%) were the main underlying conditions among CIL participants,⁹ in contrast to arthritis (76.5%), emphysema (39.6%), ulcers (35.9%), cancer (23.5%), and stroke (22.3%), which were most common in our study. It is important to recognize that the LWD program is designed for people who consider themselves to have a disability and are learning how to overcome them; in contrast, it is likely that many of the participants in our study do not consider themselves as disabled despite their activity limitations. For this reason, a study sample recruited through population screening may not have the same level of identification with and commitment to a program based on disability-related interventions.

Conclusions

Patients in this study experienced a high symptom burden, had low physical health, and took many medications. The high symptom burden suggests that there should have been many opportunities to intervene; yet dedicated, experienced study community pharmacists were not able to reduce these symptoms. They may not have been able to adapt their general MTM practices to the needs of a population with physical activity limitations, or symptom relief may not have been amenable to medication changes. The most intense CMM interventions appeared to be delivered to individuals with fewer symptoms. Common conditions such as hypertension and hyperlipidemia are often asymptomatic, and pharmacists may not be in the habit of monitoring symptoms. Future research might seek to identify which symptom management needs are most amenable to pharmacological interventions. Identifying medications or lack of medications as the cause of a symptom requires an extra cognitive step compared with more prevention-oriented problem solving, where the link between drug therapy and a health problem is supported by practice guidelines (eg, uncontrolled hypertension and osteoporosis prevention). Because of the much higher rate of symptoms and ADEs reported by people with disabilities, evidence-based practice guidelines are needed that match MTM interventions with symptomatic needs of patients. Practice guidelines, pharmacy curricula, and MTM delivery models may need to adapt to address the medication-sensitive symptoms of complex patients.

Appendix

Symptom Questionnaire

1. There are some common symptoms that people experience. We want to better understand how often they occur. As I read each statement, please say YES if you had the symptom, no matter how mild, any time in the past 4 weeks and say NO if you did not have the symptom in the past 4 weeks. [USE "DURING THE PAST 4 WEEKS" EVERY 4TH ITEM AS A REMINDER OF TIME FRAME]

	Yes	No
a. During the past 4 weeks, have you had any dizziness, lightheadedness, or problems with balance?	1	2
b. Did you have any pain?	1	2
c. Did you have any leaking of urine or problems with urinating?	1	2
d. During the past 4 weeks, did you have any problems with sleep?	1	2
e. Did you have depressed or low mood?	1	2
f. Did you have extreme fatigue or persistent tiredness?	1	2
g. During the past 4 weeks, did you have any skin problems, such as rashes, itching, burning, unusual bruising, pressure sores, or skin ulcers?	1	2
h. Did you have stomach or bowel difficulties such as nausea, vomiting, abdominal pain, constipation, or diarrhea?	1	2
i. Did you have any breathing difficulties?	1	2
j. During the past 4 weeks did you have any problems feeling confused or trouble concentrating?	1	2
k. Did you have any dry mouth or blurry vision?	1	2

2. You indicated that in the last four weeks you experienced. [LIST IN ORDER ASKED FOR QUESTION 2] [first symptom], [second symptom] . . . [last symptom].

Symptom	During the past 4 weeks, on about how many days did you have [symptom]? Was it . . .	During the past 4 weeks, on the days when [symptom] was the worst was it . . .
[first symptom]	• <7 days 1 • 7-14 days 2 • >14 days 3	• Very mild 1 • Mild 2 • Moderate 3 • Severe 4 • Or very severe 5
Add rows for each symptom reported	• <7 days 1 • 7-14 days 2 • > 14 days 3	• Very mild 1 • Mild 2 • Moderate 3 • Severe 4 • Or very severe 5

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